Conjugate-pair factor base: yield and cost

Conjugate-pair factor base in a complete toy index calculus: ≈ 36× relations per trial, 11–17× fewer trials to full rank, 2× the columns

2026-11-06. This follows `lift_dense_conjugate_pair_20261106`, which found the lines. Here they are used as a factor base in a complete index-calculus solve and compared with the standard base on the same curves, the same secret and the same targets.

The protocol was frozen before any data, at `578d0db3b0`. Amendment A1, the small-cofactor cells, was registered at `572a3df92a`, after the main run and before its own data (`PROTOCOL.md`). The instrument is `cpcost/`, std-only Rust with fresh F_{p^4} arithmetic. Report: `REPORT.pdf`.

Every statement below is an unverified assertion of this session [pcc:unverified]. The claim record is at the end.

Setup.

- $E : y^2 = (x-e_1)(x-e_2)(x-e_3)$ over $K = F_{q^4}$, so full 2-torsion; q = 13 and 101 [pcc:unverified]. P has prime order ℓ, Q = [s]P, and each trial decomposes R = [a]P + [b]Q into m = 4 factor-base points (Gaudry–Diem).
- Every relation of a trial is found exactly, by meet-in-the-middle. Relations are collected until the system has full rank. Then s is solved mod ℓ and checked.
- The arms differ only in the factor base:
  - `conj`: the first all-lift conjugate-pair line;
  - `std`: the line F_q;
  - `rand`: a random line.
- No wall-clock time is measured. All counts are exact.

Answer, small cofactor (A1: h = #E/ℓ ≤ 16, the cryptographically relevant case)

q	arm	lifted x (k)	columns D	relations / trial	trials to full rank (median)	solved
13	conj	13.0	12.8	1.811	12	16/16
13	std	6.8	6.8	0.0495	112	14/16
13	rand	6.6	6.5	0.0232	71	13/16
101	conj	101.0	101.0	1.339	92	16/16
101	std	48.7	48.7	0.0374	1518	16/16
101	rand	48.3	48.3	0.0348	1448	16/16

q	Y(conj)/Y(std)	$(k_c/k_s)^4$	ρ(conj)/ρ(std)	median T(std)/T(conj)	D(conj)/D(std)
13	36.6	13.3	19.5	11.5	1.88
101	35.8	18.5	17.2	17.2	2.07

- **The conjugate-pair base solves every curve** [pcc:unverified] (claim `PCC-20261106-328211`). It recovers the secret on 32/32 small-cofactor curves and on 32/32 curves of the main run.
- **Relations per trial are about 36× the standard base.** That is about twice the lift-density prediction $(k_c/k_s)^4 \approx 2^4$. The `std` arm itself matches the counting prediction: $(2k)^4/(24\cdot\#E)$ gives 0.036 at q = 101 against 0.037 observed.
- **Trials to full rank are 11–17× fewer.** That is net of the doubled column count. The algebra without the extra factor predicts $2^3 = 8$.
- **The extra factor of 2 is exact algebra, worked out after the data.**
  - On a conjugate-pair line, $x - e_c = u(t - \gamma)$ with $t - \gamma \in K_0$, and every element of K_0 is a square in K. So $\chi(x - e_c) = \chi(u)$ is constant.
  - In 2-descent terms, every factor-base point lies in one coset of the index-2 subgroup cut out by $(x - e_c)$ mod squares.
  - Sums of four of them always land in the subgroup itself. That subgroup contains every odd-order target, so they hit $\langle P \rangle$ twice as often as random sums.
  - The standard and random lines have no such constraint.
- **The protocol's bands were too narrow on the high side.** Y1 (yield), Y2 (per column) and Y4 (trials, q = 101: 17.2 > 16) fail as registered. Each fails because the measured advantage is larger than predicted, and the 2-descent factor accounts for it.

Main run (as frozen: any v = 1 cofactor)

q	Y(conj)/Y(std)	median T(std)/T(conj)	conj solved	std solved	rand solved
13	2.77	12.0	16/16	7/16	6/16
101	2.50	12.0	16/16	13/16	13/16

- On curves with a large cofactor (q = 13: h ≥ 84; q = 101: $h > 10^5$), $\langle P \rangle$ has only a few hundred elements [pcc:unverified]. Targets repeat and relations stop adding rank, so `std` and `rand` hit the 20,000-trial cap.
- Their pooled relations per trial is then dominated by those repeats. That is why Y1 and Y2 fail there with ratios of only 2.5–2.8. The ratio is confounded,

not small.
- Y3 (conj solves all) and Y4 (median trial ratio 12.0) pass.
- Amendment A1 removes the confound.

Cost

Derived model. Not a measurement of time. The model assumes a decomposition trial costs the same on every affine F_q-line, because the Weil-restricted summation-polynomial system has the same shape. That is not measured here.

- **Relation phase:** 11–17× fewer trials with the conjugate-pair base.
- **Linear algebra:** columns about 2×. That is about 4× for dense elimination and about 2× for sparse.
- **Brute force:** the decomposition used here costs 20,578 group additions per trial at q = 101 (`conj`) against 4,538 (`std`). That is the meet-in-the-middle over the larger factor base, not what a summation-polynomial solver would do.

Applicability (algebra plus counts).

- The family needs n even with n ≥ 4, and full 2-torsion over K. So the cofactor is divisible by 4.
- It never applies to prime-order curves, and never to odd n (EcGFp5 is n = 5).
- Among curves drawn with full 2-torsion, the share with h ≤ 16 was 16/52 at q = 13 and 16/132 at q = 101.

What this is not.

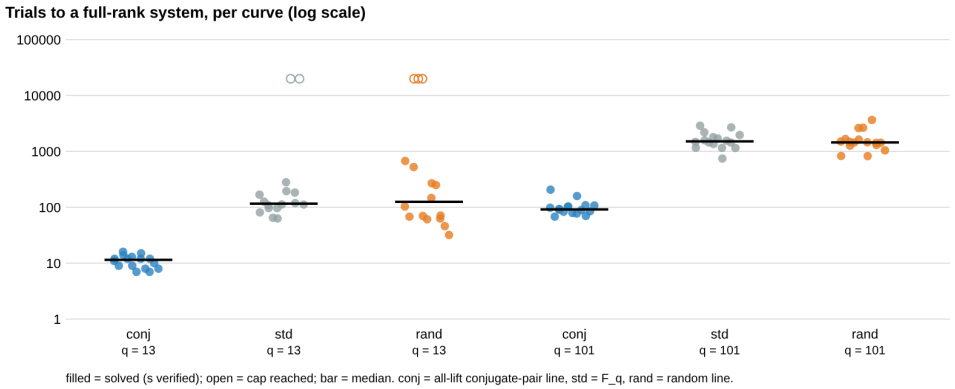
- It is not a speed measurement, and no rho comparison is made.
- It is an IC-versus-IC factor-base comparison on toy fields (q ≤ 101, n = 4).
- No leaderboard, benchmark or rho row follows from it. Every gain is a constant factor.

Registered verdicts

rule	main run	A1 (h ≤ 16)
Y1 yield ratio within [0.6, 1.6]·(k_c/k_s)⁴, rand/std in [0.5, 2]	FAIL (2.77, 2.50; confounded)	FAIL (36.6 and 35.8, above band; rand/std 0.47 at q = 13)
Y2 ρ ratio in [4.8, 12.8]	FAIL (1.75, 1.27)	FAIL (19.5, 17.2, above band)
Y3 conj solves every curve, s verified	**PASS** 32/32	**PASS** 32/32
Y4 median T(std)/T(conj) in [4, 16]	**PASS** (12.0, 12.0)	FAIL (11.5 ok, 17.2 above band)
A/A	byte-identical (`c6ba92ef…`)	see `AA.sha256`

Files

- `reports/run-20261106T0430Z/`: the main run, as frozen.
- `reports/run-20261106T0514Z/`: Amendment A1.
- Each run directory holds `rows.jsonl`, `SUMMARY.md`, `summary.json`, the figure and the A/A hashes.



trials to full rank per curve, small-cofactor cells

Reproduce

```
        cd cpcost && cargo build --release && cargo test --release &&
cd ..
        ./cpcost/target/release/cpcost run reports/run-X          #
main, ~23 min, 4 threads
        ./cpcost/target/release/cpcost run-smallh reports/run-Y   #
A1, ~11 min
        ./cpcost/target/release/cpcost summarize reports/run-Y
        ./cpcost/target/release/cpcost validate reports/run-Y
claimed.json
```

Claim record

claim	standing	content
PCC-20261106-328211	not covered (unsigned)	A1 yields, trial ratios and solves (validator-report.json, values in claimed.json)